2026 Inventors Challenge Application

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TissueTrace: A HIPAA-Compliant Smartphone Workflow for 3D Clinical Documentation in Plastic Su

5. 1-minute Elevator Pitch *

Plastic and reconstructive surgery demands the highest imaging precision of any surgical specialty. Yet our standard of care, the 2D smartphone photograph stored as a JPEG. We are building an end-to-end platform that supplements static photography with interactive, photorealistic 3D documentation using the smartphone already in the surgeon's pocket.

Today, consumer apps can turn a short video into a 3D reconstructio the same technology behind Apple’s Persona avatars on the Vision Pr not HIPAA-compliant, do not connect to the medical record, and are TissueTrace fills this gap. A clinician captures a 15-second video orbit of the anatomy, and our pipeline produces a reconstruction that preserves how tissue actually appears under light. This allows surgeons to examine the way perfusion, bruising, and color appear across different viewing angles, rather than relying on a flat image.

What is new here is not the 3D technology, but the clinical workflow integration. TissueTrace combines HIPAA-compliant capture, automated 3D processing, and EHR-linked output in a single application. The result is stored in a secure repository linked directly to the patient’s chart. The electronic health record remains the system of record; TissueTrace is simply the tool that makes clinically useful 3D data accessible within it.

6. Need or Problem Your Product Addresses *

Plastic surgery depends uniquely on precision imaging. Our clinical decision-making, operative planning, outcomes assessment, and patient communication rely on accurate visual documentation of surface anatomy. Yet the standard of care, a 2D smartphone photograph stored as a JPEG, has not fundamentally changed in decades.

Surveys consistently demonstrate that smartphone photography dom
documentation in our field. Between 96–99% of plastic surgeons and
for clinical photos (Lam et al. 2019; Veith et al. 2022; Grow et al. 20
HIPAA-compliant storage (Veith et al. 2022), and 58% of surgical residents report sharing protected health information via text messaging in violation of HIPAA (McKnight Franko 2016). The American Society of Plastic Surgeons explicitly discourages the use of insecure email or text message communication with patients (ASPS Statement on Telehealth 2020). These 2D images have critical limitations that directly affect clinical care:

Inadequate for surgical planning and assessment: A single 2D image cannot capture the contour, projection, symmetry, and surface quality needed to plan or evaluate procedures such as tissue expansion, free flap reconstruction, breast reconstruction, or scar revision. Wounds with depth—including lacerations, expander pockets, and flap insets—have surfaces facing multiple directions that a handful of photos cannot adequately document.

Inconsistent and non-retroactively standardizable: OR photos, clinic photos, and home photos are compared longitudinally as if they were equivalent. There is no practical way to standardize lighting or perspective in existing 2D images.

No quantitative measurement capability: Surface area, volume, curvature, and deformation cannot be derived from 2D images. Existing 3D systems such as Vectra and 3dMD are cost-prohibitive, require dedicated hardware, and are primarily adopted for aesthetic consultation simulation (Rudy et al. 2024; Almadori et al. 2022).

Incomplete documentation: A single-angle 2D photograph is an inherently incomplete record of three-dimensional anatomy. A 3D reconstruction captures more of the clinical reality, potentially supporting more thorough documentation of preoperative conditions and postoperative outcomes.

Not prepared for emerging computational applications: Machine learning models for flap perfusion monitoring, wound classification, and outcome prediction are advancing rapidly, but they are bottlenecked by inconsistent 2D training data. Virtual surgical planning is standard for craniofacial bone but lacks patient-specific soft tissue surface data. As these technologies mature, the field will need high-fidelity 3D documentation infrastructure that does not yet exist.

Dead-end data format: A JPEG stored in the EHR can never yield more information than it captured. It cannot be re-examined from new angles, reprocessed with future algorithms, or mined for 3D features. It is a static endpoint, not a foundation for future innovation.

7. Short Summary of What the Product Does *

The product is an end-to-end platform. A clinician captures a short smartphone video of approximately 30 seconds orbiting the clinical area of interest during routine encounters. Our processing pipeline generates a photorealistic, interactive 3D reconstruction that is hosted on TissueTrace's HIPAA-compliant media server and linked to the patient's chart in the EHR via standard health IT integration protocols such as SMART on FHIR. A link appears in the patient's chart; when any authorized provider clicks it, the web-based interact [illegible] 3D model for that encounter. This is the same integration architectur [illegible] systems (PACS) and by other clinical media platforms that host conte [illegible] EHR systems do not natively support 3D file formats, and this architecture solves that limitation transparently.

Technical pipeline: The video is processed through a three-stage computer vision pipeline. First, Structure from Motion, often implemented via COLMAP, estimates the camera pose for each video frame and produces a sparse 3D point cloud through feature matching and bundle adjustment. Second, the sparse point cloud initializes a 3D Gaussian Splatting (3DGS) reconstruction, which is a real-time radiance field rendering technique introduced at SIGGRAPH 2023. This technique represents the scene as a collection of volumetric Gaussian primitives encoding position, orientation, opacity, and view-dependent radiance via spherical harmonics. Third, emerging mesh-in-the-loop extraction methods, such as MILo (Guédon et al., 2025), extract metrically accurate triangulated meshes from the optimized Gaussians while guiding the optimization toward surface-consistent configurations.

The platform would support:
- Interactive 3D clinical documentation: Any provider examines anatomy from any angle, at any time, from any device. It supports assessment of symmetry, contour, wound healing, flap viability, and scar quality. These are applications where plastic surgery uniquely requires precision that 2D photos cannot provide.

- Remote 3D consultation: A consulting surgeon explores anatomy interactively from their preferred assessment angles rather than interpreting a single flat image. This enhances tumor boards, multidisciplinary conferences, and telemedicine encounters.

- Longitudinal 3D tracking: Serial reconstructions across clinic visits create a 3D timeline of tissue expansion progression, flap incorporation, wound healing trajectory, or scar maturation, all viewable interactively at each time point.

- Quantitative surface analysis: Mesh extraction from the archived 3DGS reconstructions enables measurement of surface area, volume, curvature, and deformation. This extends into biomechanical modeling, such as finite element analysis, for soft tissue simulation. This also extends the digital twin paradigm, currently standard for craniofacial bone, into the soft tissue domain.

- Resident and medical education: A library of interactive 3D clinical cases is categorically richer for teaching surgical anatomy and technique than a slideshow of 2D photographs.

Future-proof data infrastructure: Because 3DGS encodes richer appearance and geometric information than any 2D or mesh format, the archived data can support applications that do not yet exist in routine clinical use but are emerging rapidly. These include standardized relighting, which allows for retroactively normalizing illumination across time points for unbiased outcomes comparison, as well as high-fidelity training data for machine learning models and deformable soft tissue simulation for virtual surgical planning. The goal is not to predetermine how this data will be used, but to ensure that when these tools arrive, the imaging infrastructure is ready for them. A JPEG stored today will never support these applications, whereas a 3DGS reconstruction stored today can.

8. Product Stage of Development *

Concept stage with working proofs-of-concept using off-the-shelf components.

The underlying 3D Gaussian splatting technology is mature:
- Multiple consumer apps producing high-quality reconstructions from smartphone video.
- 3DGS produces photorealistic reconstructions with state-of-the-art visual fidelity (Kerbl et al., 2023), with recent forensic studies demonstrating millimeter-level measurement accuracy from smartphone-captured 3DGS reconstructions (Cho and Woo, 2026).
- Smartphone-based 3D scanning has been validated against the Canfield Vectra H2 for breast surface measurements with sub-millimeter discrepancies (Rudy et al., 2024).
- Published demonstrations exist in surgical scene reconstruction (EndoGaussian, SurgicalGaussian, Free-SurGS), wound assessment (Wound3DAssist), and teledermatology. These are research publications demonstrating technical feasibility, not integrated clinical products.
- The glTF 2.0 KHR_gaussian_splatting extension was ratified by the Khronos Group in 2026, providing a standard interoperable file format.
- EHR integration via SMART on FHIR is a well-established pathway used by clinical imaging and documentation platforms across specialties.

What does not yet exist is the integrated 3D clinical product: HIPAA-compliant 3D media hosting, automated video-to-splat processing, mesh extraction tools, and the EHR integration layer, all packaged in a surgeon-facing interface.

To support future clinical implementation, we have planned a validation study to compare 3D Gaussian splat-derived meshes against our lab group's previously established clinical 3D imaging (Vectra H2) and an established multi-view stereo pipeline in a porcine tissue expansion model. This proposed work aims to inform acquisition protocols, quality thresholds, and appropriate clinical use cases. While intended to proceed independently of the challenge outcome, prize funding would significantly accelerate the platform build and the integration of this validation data.

9. Competitors/Competition in the Market *

Canfield Scientific (Vectra H2): the industry standard for 3D clinical photography. Uses stereophotogrammetry with proprietary hardware costing $20,000–$50,000+. Primarily deployed for aesthetic simulation such as breast augmentation and rhinoplasty. Not typically used for routine reconstructive documentation or longitudinal wound and tissue tracking.

eKare (inSight): an FDA-registered wound measurement device with
chronic wound care with automated area, volume, and depth measu
on wound metrics rather than broader reconstructive surgery docum

ImageAssist (Golinko et al.,): a smartphone app that standardizes 2D clinical photography and integrates with the EMR via SMART on FHIR; 3,400+ photographs captured in 800+ encounters in its first two years. ImageAssist addresses the important problem of photographic standardization and EHR integration for clinical images. TissueTrace addresses a different clinical need by capturing and integrating 3D surface geometry for cases where quantitative measurement and interactive spatial examination are valuable. The two tools are complementary and could coexist within the same department, serving different documentation needs.

Consumer 3D scanning apps (Scaniverse, Polycam, KIRI Engine, Luma AI): produce high-quality 3D Gaussian splats from smartphone video. Not HIPAA-compliant, no clinical workflow, no EHR integration, and not designed or validated for medical use.

Our differentiation: No existing clinical product integrates (1) smartphone-based capture requiring zero additional hardware, (2) 3DGS reconstruction preserving photorealistic, view-dependent tissue appearance, (3) integrated mesh extraction enabling quantitative geometric analysis, (4) HIPAA-compliant hosting with EHR integration via standard health IT protocols, (5) a focus on the reconstructive surgery use cases that uniquely demand precision 3D documentation, and (6) a future-proof data format that can support emerging computational applications.

10. Team Information (please describe who is on your team if applicable)

Christian Arcelona is an M.D. candidate at Chicago Medical School (Rosalind Franklin University) and a research fellow in the Gosain Plastic Surgery & Craniofacial Biology Laboratory at Ann & Robert H. Lurie Children's Hospital of Chicago. He holds an A.A.S. in Mechanical Engineering and a B.S. in Biological Sciences with minors in Chemistry and Mathematics. His engineering and programming background is directly relevant to TissueTrace. He brings hands-on experience bridging software development with clinical plastic surgery workflows [illegible]

Ethan D'Silva, MD is a General Surgery resident at the University of [illegible] Pediatric Plastic Surgery Research Scholar with Dr. Gosain at Ann & Robert H. Lurie Children's Hospital of Chicago. He earned a BSA in Neuroscience from the University of Texas and an MD from Baylor College of Medicine. Dr. D'Silva has experience building and coordinating international, multicenter clinical databases, developing automated data collection tools, and applying machine learning to predict clinical outcomes. He has also led implementation of surgical training infrastructure in a low-resource settings. For TissueTrace his experience in multicenter study coordination, scalable data pipelines, and outcomes research will support clinical workflow design and pilot evaluation.

Joanna K. Ledwon, PhD is a Research Assistant Professor of Surgery at Northwestern University Feinberg School of Medicine and the Stanley Manne Research Institute at Lurie Children's Hospital. Her research focuses on the molecular and cellular mechanisms of mechanically induced skin growth, with extensive experience in the porcine tissue expansion model central to TissueTrace's planned validation study. Dr. Ledwon has collaborated with Dr. Tepole's group to build computational frameworks coupling biological data to mechanical models, including isogeometric analysis of skin deformation and Bayesian calibration of growth laws. She will oversee the experimental validation comparing 3D Gaussian splat-derived meshes against the Vectra H2 and established multi-view stereo pipelines.

Adrian Buganza Tepole, PhD is an Associate Professor of Mechanical Engineering at Columbia University and recipient of the ASME Y.C. Fung Young Investigator Medal. His lab developed the multi-view stereo and isogeometric analysis pipeline (Buganza Tepole et al., 2015) that serves as a reference benchmark for TissueTrace's validation study. With over 14 years of collaboration with Dr. Gosain's team on computational modeling of skin mechanics, his expertise in surface reconstruction, constitutive modeling, and patient-specific digital twins directly supports the development of TissueTrace's mesh extraction and quantitative measurement tools.

Arun K. Gosain, MD is Head of the Division of Pediatric Plastic Surgery at Lurie Children's Hospital and Professor of Surgery at Northwestern University, with over 300 peer-reviewed publications and more than two decades of continuous NIH-funded research. He has served as Chair of the Board of Trustees of the American Society of Plastic Surgeons and President of the American Council of Academic Plastic Surgeons. Dr. Gosain provides senior clinical and scientific mentorship, access to the porcine tissue expansion model and Vectra H2 system for validation, and the institutional infrastructure for future clinical pilot testing.

Additional team members include research fellows and lab technologists.

11. Any Other Information You Would Like to Add

Building on established methodological foundations: The computer vision principles underlying our platform have a direct lineage in plastic surgery research. Multi-View Stereo reconstruction from photographs has been validated for characterizing porcine skin [illegible] reconstruction errors of 1-4.7% (Buganza Tepole et al., 2015). Our p[illegible] fundamental principles with modern tools. COLMAP replaces legacy s[illegible] traditional textured mesh representation with a format that preserves view-dependent appearance; and emerging mesh extraction methods produce clean meshes that can feed directly into biomechanical modeling pipelines.

Why 3D Gaussian Splatting, and why now: 3DGS was introduced at SIGGRAPH 2023 and has rapidly become one of the most active areas in computer vision. In the surgical domain, deformable 3DGS has been applied to endoscopic tissue reconstruction, and relightable 3DGS has been demonstrated for physically-based tissue rendering. Mesh extraction from Gaussian representations has advanced rapidly in 2025 and 2026, making 3DGS increasingly viable for both visual documentation and quantitative geometric analysis. Consumer smartphone apps already support 3DGS capture, demonstrating that the capture pipeline is hardware-accessible today. The technical components are maturing rapidly; what is missing is the clinical platform, workflow integration, and validation for plastic surgery.

Digital twinning and virtual surgical planning: Virtual surgical planning is already standard of care for craniofacial bone work using CT-derived models. However, soft tissue simulation remains limited by the lack of patient-specific surface geometry captured at scale. TissueTrace-derived meshes of a patient’s actual tissue, captured longitudinally, could serve as the foundation for finite element-based predictive models of tissue behavior, flap behavior, and aesthetic contour outcomes. This extends the digital twin paradigm from bone into the soft tissue domain.

Preparing for the future of clinical AI: Machine learning applications in plastic surgery are advancing rapidly. These models require large, standardized, high-fidelity imaging datasets that do not currently exist. By adopting TissueTrace documentation now, the field begins building a retrospective archive of rich 3D clinical data that can support these applications as they mature. The goal is not to predetermine how this data will be used, but to ensure that when these tools arrive, the imaging infrastructure is ready for them.

Impact on daily clinical workflow: The behavioral change required for adoption is minimal. Nearly all plastic surgery trainees already use their smartphones for clinical photos. We are asking them to capture a 30-second video orbit instead of snapping 3 to 5 static photographs. The capture takes the same amount of time, processing is automatic, and the result is a richer clinical record that supports better surgical planning, consultations, and documentation without adding time or cost to the clinical encounter.

12. Pitch Deck: Once this form is submitted, please email your Pitch Deck directly to jcross@plasticsurgery.org

Thank you for your time and consideration.